

Full paper

Phytic acid derived LiFePO₄ beyond theoretical capacity as high-energy density cathode for lithium ion battery

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ABSTRACT

A glucose-derived carbon-encapsulated (GC) LiFePO₄ (LFP) with internal carbon sheets (IC) composite, GC/IC/LFP, is synthesized by using phytic acid (PhyA) as a special phosphorus source. PhyA plays an important role as a structure-director, complexing agent and antioxidant, and the resulted IC contributes to improving electrochemical properties. The special morphology and microstructure of precursors and final materials are characterized and discussed in detail. The prepared GC/IC/LFP shows not only an excellent cycle performance, but an ultra-high specific capacity of 192 mA h g⁻¹ at 0.1 C, which exceeds the theoretical capacity of LFP. A remarkable rate performance of 140 mA h g⁻¹ at 10 C was also achieved. The energy efficiency of GC/IC/LFP can reach nearly 96% at 0.1 C.

1. Introduction

An increasing attention of the world has been taken to the electric vehicles (EVs), hybrid electric vehicles (HEVs) and plug-in hybrid EVs (PEVs) equipped with high-power and high-capacity Li ion batteries (LIBs) [1,2]. As one of the most promising cathode materials for LIBs, olivine structured LiFePO₄ (LFP) was first reported by Goodenough in 1997 [3], which has attracted immense interest due to its large-scale and high-rate applications [4]. It has been reported that LFP as cathode materials for LIBs possesses many significant advantages, including relative high capacity, high safety, intrinsic stability, acceptable operating voltage (3.4 V vs. Li⁺/Li), environmental compatibility and low cost [5–7].

However, the theoretical capacity of LFP is lower (170 mA h g⁻¹) than that of some other common cathodes and the tap density is also low with a range of 1.1–1.3 g mL⁻¹. Moreover, the sluggish kinetics of Li ion and electron transport is another barrier in its development [8]. Some lithium ions cannot be fully extracted from the ordered-olivine structure due to the low ionic mobility, which in turn causes some capacity loss, resulting an even lower commercial capacity of LFP (~150 mA h g⁻¹). As a result, most applications of LFP are now only in

stationary energy storage such as renewable energy and smart grids, as well as on-board energy storage such as HEVs, EVs and PEVs, instead of portable electronic devices. Enormous efforts have been made to conquer the drawback by coating the cathode with electronically conductive carbon or conductive polymers [9–13], doping with foreign metals [14–16], and reducing particle size [17–19]. It is noteworthy that Hu et al. [13] applied the electrochemically exfoliated few-layer graphene flakes (EG flakes) to wrap the commercially available cathode material LFP, the capacity of which is beyond the theoretical 170 mA h g⁻¹. The extra capacity is mainly ascribed to the reversible redox reaction between the lithium ions of the electrolyte and the EG flakes.

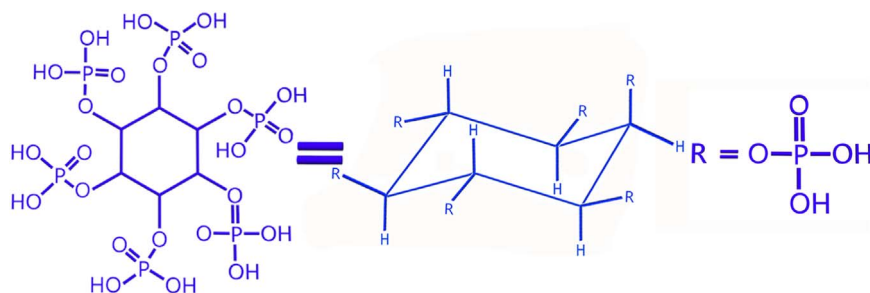
Recently, an increasing number of biomass materials have been used to prepare cathode and anode materials for LIBs due to their environmental friendliness, abundance and sustainability [20–22]. Especially, the internal carbon source of biomass materials can usually influence their final morphologies and properties [23]. Phytic acid (PhyA, Scheme 1), with the chemical formula of C₆H₆(H₂PO₄)₆ known as inositol hexakisphosphate or phytate in salt form, is the principal storage form of phosphorus [24]. It is a natural and nontoxic, organic large molecular compound which presents in most of legumes, includ-

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Scheme 1. Structure of phytic acid.

ing soy beans nuts, and corns [25,26]. However, as an anti-nutritional factor, PhyA restricts the application of legumes to food or fodder [27]. Therefore, the exploitation and utilization of PhyA is necessary. The PhyA molecule contains six carbon atoms and six phosphate groups [28,29]. The special structure of PhyA enables it to chelate with positively charged multivalent cations such as Fe^{2+} , Fe^{3+} , Ca^{2+} , Mg^{2+} , and Zn^{2+} due to its strong negative charge on the surface [30]. These complexes [31] are soluble under acidic aqueous solution but precipitate at neutral pH [32]. Herein, PhyA is utilized to synthesize LFP as peculiar phosphorus source by hydrothermal/solvothermal synthesis methods. The PhyA derived carbon sheets in LFP (defined as IC) can be used as Li^+ storage sites. Moreover, its internal carbon source and symmetric structure might improve the performance and structure of LFP.

In this paper, LFP precursor with internal carbon sheets was synthesized with a facile hydrothermal method employing PhyA as a phosphorus source, antioxidant and structure-directing agent. The effect of PhyA and the change of composite structure were discussed in detail. And then, the precursor composite was coated with the glucose-derived external carbons by the following calcination. The as-obtained composite shows a special microstructure and an excellent electrochemical performance. More significantly, the composite has a higher discharge capacity than the theoretical capacity of LFP.

2. Experimental section

2.1. Synthesis of IC/LFP

All chemicals were used without any further purification. PhyA (Eastern Co., Chengdu, 70%) and $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ (Kermel Chemical Reagent Co., Ltd., Tianjin, 99%) were first dissolved in a solution mixture of ethylene glycol (EG) and deionized water to form a yellowish-brown transparent solution. $\text{LiOH} \cdot \text{H}_2\text{O}$ (Kelong Chemical Reagent Factory, Chengdu, 99%) solution was added slowly to the above solution. The molar ratio of $\text{FeSO}_4/\text{PhyA}/\text{LiOH}$ was 5:1:18 in the starting solution and pH of the solution was adjusted to ~7.2. After bubbling with high-pure Ar for 0.5 h, the mixture was quickly transferred to a stainless steel autoclave with a mechanical stirrer, which was heated at 200 °C for 2 h, and then cooled to room temperature naturally.

The as-prepared precipitant was collected by centrifuging, washed with absolute alcohol for only one time, and dried at 60 °C under vacuum for 8 h. Then the composite was sintered at 750 °C for 2 h in a horizontal tube furnace under a high-purity Ar atmosphere. Finally, IC/LFP was obtained.

2.2. Synthesis of GC/IC/LFP

In order to further improve the conductivity of products, the glucose was doped into the precursors before calcination. The glucose was weighed in stoichiometric amounts and was then mixed thoroughly with the precursors. The weight ratio of glucose to precursor was 1: 9. In the same way of calcination, GC/IC/LFP was obtained.

2.3. Synthesis of LFP and GC/LFP

As a reference, LFP and GC/LFP were also prepared by using phosphoric acid (PA) as phosphorus source. All of procedures and reagents are the same with the above way except the choice of phosphorus sources.

2.4. Synthesis of C/LFP/ECGO

As a reference, a carbon coated LFP composite based on electrochemically cleaved graphite oxides (ECGO) was prepared, which has been reported in one of our previous work by using PA [33]. The C/LFP/ECGO electrode achieved a higher specific capacity and a better rate performance.

2.5. General characterizations

The obtained powders were subjected to various characterization techniques. X-ray photoelectron spectra (XPS) were acquired on a Kratos XSAM 800 spectrometer (Manchester, UK) with a MgK α X-ray (1253.6 eV) excitation source running at 15 kV, with a hemispherical electron energy analyzer and a multichannel detector. The TG-DTG analysis was performed using a SDT Q600 thermal analyzer (New Castle, USA). Raman spectra were detected on a Confocal LabRAM HR800 spectrometer, HORIBA Jobin Yvon (Paris, France) to analyze the disorder degree of porous carbon materials (ratio of D band and G band of the carbon) using 532 nm excitation laser. Field emission scanning electron microscopy (SEM) images were obtained with a Hitachi S4800 scanning electron microscope (Tokyo, Japan). Transmission electron microscopy (TEM) images were acquired on a FEI Tecnai G2 20 TEM (Hillsboro, OR, USA) operating at 200 kV. The crystal structure and phase purity of the obtained samples were analyzed using a Tongda TD-3500 X-ray powder diffractometer (Dandong, China) with CuK α radiation ($\lambda=0.15148$ nm) operating at 30.0 kV and 20.0 mA. Elemental analysis was done by using EuroEA3000 analyzer (Foggia, Italy) Analyzer. Nitrogen absorption isotherm was measured with Micromeritics Tristar 3020 automatic analyzer (USA). Inductively coupled plasma mass spectrometry (ICP-MS) was performed using a VG PQExCell analyzer (Boston, USA).

2.6. Electrochemical tests

The electrochemical behaviors of IC/LFP, GC/IC/LFP and GC/LFP cathode material were evaluated by button cell (CR2032). With N-methyl pyrrolidone (NMP) served as the solvent on an aluminum foil, the working electrode was made by pressing a mixture of 85 wt% active material and 8 wt% acetylene black, with 7 wt% polyvinylidene fluoride (PVDF) as binder. The lithium plates were used as the counter electrode. The electrolyte was prepared by dissolving 1 M LiPF_6 in a mixture of dimethyl carbonate (DMC) and ethylene carbonate (EC) (1:1 in volume). A Celgard 2400 microporous film was used as a separator. The loading mass of the prepared electrodes is about 2.8 mg cm^{-2} and the porosity for the electrodes is about 26%. These

coin-type anodes were assembled in an argon-filled Delix LS800S glove box (Sichuan, China). The cells were galvanostatically charged and discharged between 2.2 and 4.2 V vs. Li⁺/Li on a Neware CT-3008W battery cycle (Guangdong, China). Electrochemical impedance spectroscopy (EIS) was carried out in the frequency range of 0.01–100 kHz at a charged stage with an applied amplitude of 5 mV. The cyclic voltammograms (CV) was obtained at a scanning rate of 0.1 mV s⁻¹ on an Autolab PGSTAT 302 electrochemical workstation (Eco Chemie B.V., Amsterdam, and the Netherlands).

3. Results and discussion

After the hydrothermal process, the dark green precursor can be obtained, which could be turned into the final dark product after calcinations. To compare the structure of precursor and final product, we tested these two materials with XPS, Raman spectra and TGA. The structure and effect of GCs have been discussed in previous reports [33]. Therefore, in these characterizations, only IC/LFP, without the addition of glucose, is tested in order to confirm the role of IC.

XPS was carried out to examine the composition and the chemical state of the atoms in the products before calcination and after calcination without the addition of glucose (IC/LFP). The XPS spectrum in Fig. 1a shows the C1s spectrum of precursor before calcining process, which possesses three main peaks that correspond to functional groups C-C, C-O and C=O, respectively [33]. The existence of C-O (Fig. 1a) indicates that the precursors are phytates and the structure of phytates is not disrupted during the hydrothermal process. However, the functional group C-O at 285.950 eV disappears after calcining at 750 °C (Fig. 1c), indicating the rupture of the chemical bond C-O in phytates, which can also be proved by the O1s spectrum. In Fig. 1b, O1s spectrum shows two main peaks, which can be ascribed to the functional groups O-C and O-P, respectively [34,35]. The chemical bond O-C also ruptures after calcining according to the absence of O-C peak (Fig. 1d). Therefore, the phytates have eventually decomposed into phosphates. In conclusion, results from XPS show that the phytates molecule in precursors survives during the hydrothermal process but decomposes into phosphates during the calcination

process. During the decomposition process, the six-member carbon ring of PhyA would be reconstructed as the internal carbon (IC) sheets.

Results above were further verified by Raman spectrum and thermogravimetric analysis (TGA). The Raman spectra of precursor and final product are collected by using an excitation laser at 532.17 nm (Fig. 2). The assignments of peaks are summarized and consistent with the literatures [36–38]. The bands at 989 and 1028 cm⁻¹ in Fig. 2a are ascribed to the functional groups P-O-C and C-C-C [39], which indicates that the structure of phytate is stable during hydrothermal process. Nevertheless, the structure of the final product changed dramatically after calcining (Fig. 2b). The two bands at 989 and 1028 cm⁻¹ decrease severely and the band at around 961 cm⁻¹ emerge, indicating that the phytates have been decomposed into phosphates during the calcination process. In addition, another pair of peaks rise in Fig. 2b, corresponding to the characteristic peaks of carbon, providing a strong evidence of the presence of the PhyA-derived ICs. The strong band at 1337 cm⁻¹, which is known as the disorder-induced phonon mode (D-band), originated from the disorder-induced features of carbon and structural defects [40]. Moreover, the band observed at 1607 cm⁻¹, known as the graphite band (G-band), is the characteristic peak of carbon materials with a high degree of ordered structure [41,42]. The I_D/I_G value (the peak intensity ratio between the 1337 and 1607 cm⁻¹ peaks) generally provides a useful index for comparing the degree of crystallinity of various carbon materials. The large I_D/I_G ratio in Fig. 2b indicates a high degree of disorder due to the presence of defects on ICs induced by high content of oxygen, which will be verified by the following analysis of ICs below.

Fig. 2c and d show the TGA of the precursors of LFP (using PA as phosphorus source in the process of preparation) and IC/LFP. As seen in Fig. 2c, the weight loss that appears under 200 °C is ascribed to the loss of free water in both phosphate and phytate. After that, two obvious weight loss stages exist in the TGA curve of phosphate. From TGA curve of phytate, an unobvious weight loss stage is observed. The differential curves of two TGA profiles are derived to demonstrate their difference. In Fig. 2d, two peaks at around 300 °C and 460 °C exist in both differential curves, corresponding to the loss of unbound water and constitution water. The slight offset of two differential curves can

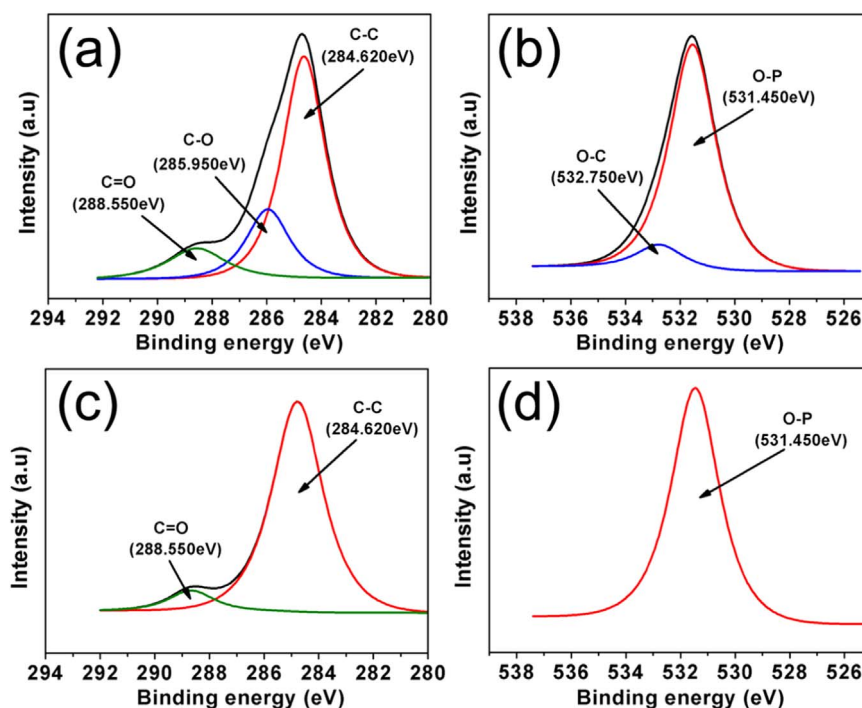


Fig. 1. XPS C1s spectrum (a) and O1s spectrum (b) of precursors before calcination and C1s spectrum (c) and O1s spectrum (d) of final products after calcination without addition of glucose.

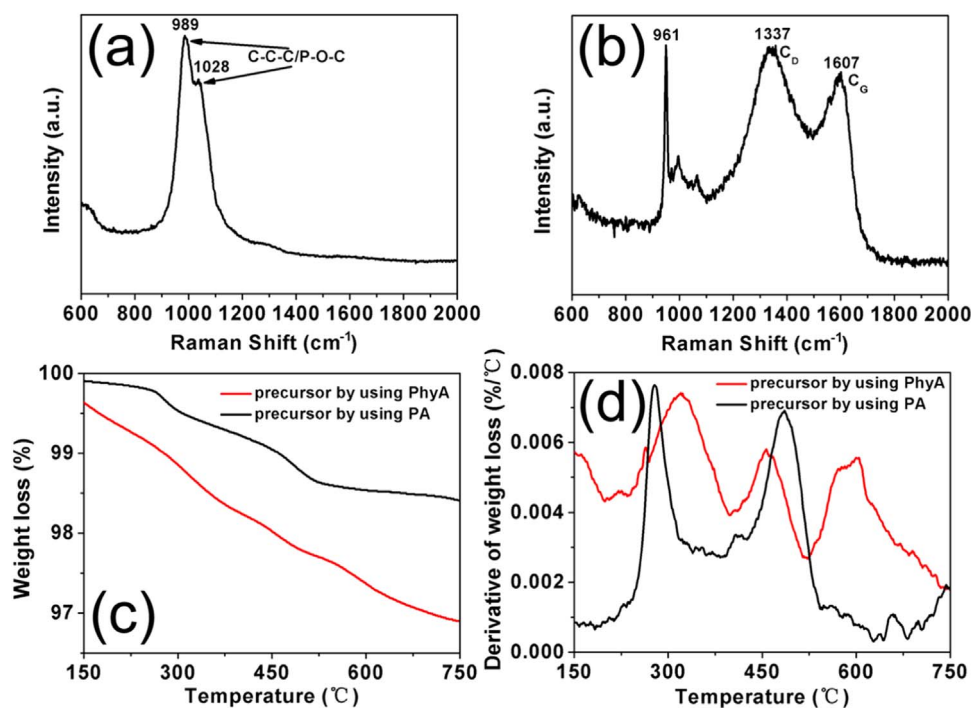


Fig. 2. Raman spectra of precursor (a) and final product (b) without addition of glucose. (c) TGA of two different precursor by using PhyA and PA as phosphorus source at a heating rate of 8 °C min⁻¹ in nitrogen at a 60 mL min⁻¹ flow rate without addition of glucose. (d) The differential curves of TG profiles in (c).

be addressed to the difference of structures of two materials. Furthermore, the phytate curve has an additional peak at around 600 °C, which is ascribed to the oxidation of organic matter [43]. The thermogravimetric analysis indicates that the phytates could start to decompose into the internal carbons and phosphates (IC/LFP) at about 600 °C. In addition, it can also be proved that the phytate is not ruptured within 200 °C.

Based on the discussed above, the possible schematic illustration of synthesis of IC/LFP is proposed as shown in Figs. S1 and S2. The structure of phytates in precursors is stable during the hydrothermal process. After calcination, the phytates decompose into phosphates. During the decomposition process, the six-member carbon rings of PhyA are reconstructed into ICs which are highly disordered and rich in defects.

To investigate GCs and ICs individually, the LFP in GC/LFP and IC/LFP were dissolved through an acidizing process. The GC/LFP and IC/LFP powders were added separately into the hydrochloric acid solutions (3 mol L⁻¹) under magnetic stirring. Then the solutions were heated to boiling for 5 min and were cooled to room temperature naturally. Finally, the ICs and GCs can be obtained after the solutions were centrifuged and the participants washed over 3 times. The ICs and GCs were characterized and their structures as well as properties have been compared as below.

Fig. 3a shows the X-ray diffraction (XRD) patterns of GCs and ICs. The patterns of GCs and ICs show two broad diffraction peaks around 26° and 44° that are indexed as (002) and (100) of the pseudographitic domains [44]. To understand the graphitization degrees of the materials, Dahn et al. used the empirical parameter (*R*), which is defined as the ratio of height of the (002) Bragg peak to the background [45]. As shown in Fig. 3a, the GCs delivers an obvious higher *R* value than ICs, indicating it possess a higher degree of graphitization and is expected to have a better conductivity [46]. In this work, GCs was employed to enhance the conductivity of IC/LFP.

Raman spectra are sensitive to the slight structural changes and have been widely used to characterize carbon materials. As shown in Fig. 3b, both of GCs and ICs exhibit a broad disorder-induced D-bands at about 1340 cm⁻¹ and an in-plane vibration G-bands at about 1591 cm⁻¹. The value of the integral intensity of the D- and G-band could be obtained by fitting the spectra and *I_D/I_G* was employed to index the degree of disorder. The *I_D/I_G* intensity of ICs is 0.96, higher than that of GCs (0.91). This represents a higher degree of disorder, more edges, and more other defects. The defects on ICs could provide some binding sites for oxygen groups, which would improve the electrochemical performance for GC/IC/LFP. Nitrogen adsorption-desorption isotherms were employed to characterize the porous

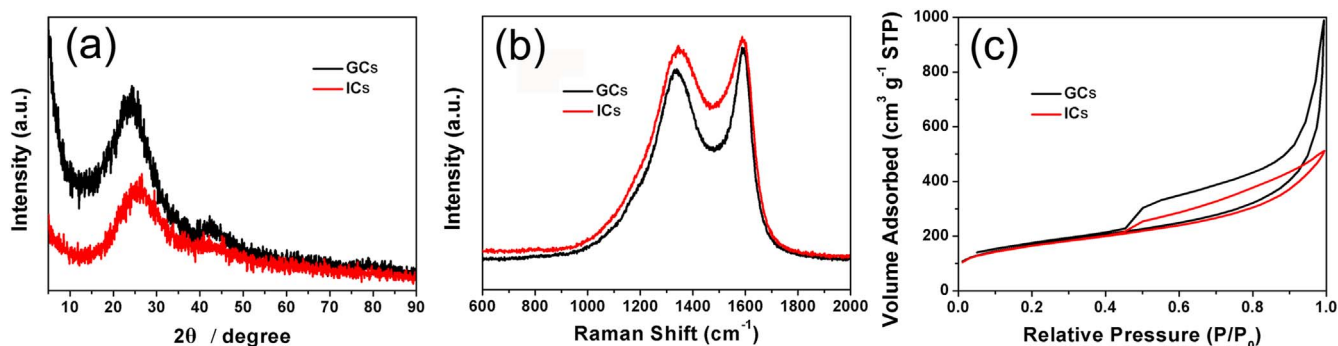


Fig. 3. (a) XRD patterns, (b) Raman spectra and (c) Nitrogen adsorption-desorption isotherms for GCs and ICs.

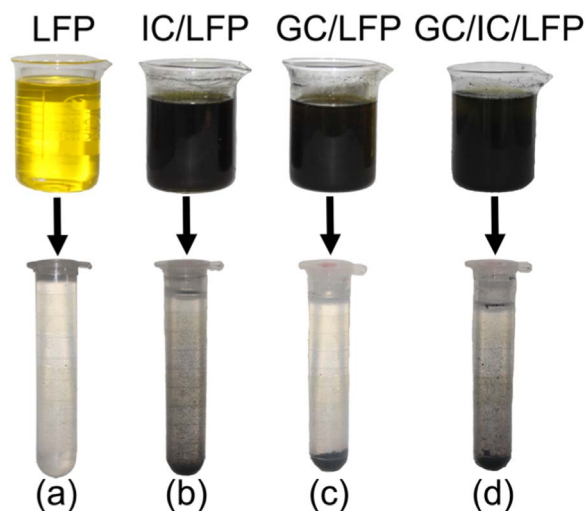


Fig. 4. The photographs of (a) LFP, (b) IC/LFP, (c) GC/LFP and (d) GC/IC/LFP after acidizing.

structures of GCs and ICs (Fig. 3c). Both GCs and ICs deliver a type IV N_2 -adsorption isotherm with a H3/H4 hysteresis loop, indicating the presence of some mesopores. The Brunauer-Emmett-Teller (BET) specific surface area of ICs is $586.6 \text{ m}^2 \text{ g}^{-1}$, close to that of GCs ($596.1 \text{ m}^2 \text{ g}^{-1}$).

The photographs of four materials, LFP, IC/LFP, GC/LFP and GC/IC/LFP, after being acidized are shown in Fig. 4. It is obvious that there is no existence of carbon in LFP (Fig. 4a), by using PA without addition of glucose. By contrast, the ICs indeed exist in IC/LFP (Fig. 4b), synthesized by PhyA without the addition of glucose. This result demonstrates that PhyA indeed can induce ICs in LFP. After doping of the glucose into LFP, the GCs can be obtained (Fig. 4c). And the result of GC/IC/LFP is also shown in Fig. 4d. In addition, the resulted precipitants were washed and separated by low-speed centrifugation. As shown in Fig. 4, ICs are suspended in the deionized water for a few seconds while GCs sink to the bottom of the centrifuge tube rapidly. Fig. 4d shows the mixture of ICs and GCs. Compared with GCs, the ICs demonstrated a better hydrophilicity, due to the abundant of oxygen groups, which is also confirmed by XPS. As shown in Fig. 3S, the peaks for XPS C1s binding at 285.78 eV (C–O), 287.642 eV (C=O) and 289.598 eV (O=C–OH) suggests the presence of densely bonded oxygen-containing functional groups. The amount of oxygen is too low to be detected by XPS for GCs. For carbon materials, the introduction of surface functional groups, like oxygen, is a promising method to increase the energy density of electrodes [47].

The comparison of morphology and structure among ICs, GCs and GC/IC is shown in Fig. 5. The larger cavities in the images are due to the acidizing and removing of LFP. It is obvious that ICs (Fig. 5a and b) possesses a hierarchical three-dimension network structure, which can promote the mobility of lithium-ions. From Fig. 5b, the ICs show a thin and fine sheet or strip structure. In addition, there are many micropores came from the highly disordered aggregation of ICs (the red dotted circle in Fig. 5c), which is also beneficial to the transportation of lithium-ions. Therefore, lots of defect sites would appear on ICs, which could induce densely active sites for oxygenic groups and therefore provides the possibility of extra capacity. For comparison, the morphology and structure of GCs are shown in Fig. 5d and e. From the vacancies where the LFP are dissolved, it can be observed that the GCs coating on the surface of LFP, which can improve the conductivity of LFP. After GC/IC/LFP was acidized, the GC/IC was obtained. As shown in Fig. 5g and h, the ICs are wrapped by GCs thus can hardly be located in the SEM/TEM images. Fig. 5i shows the micropores and large cavities, proving the coexistence of GCs and ICs.

As discussed above, the porous ICs possess lots of defects and O-

containing groups, which could provide extra sites for Li-ion storage as electrode materials for LIBs. The hydrophilicity of ICs is excellent, while the conductivity of them is not as good as that of GCs. Therefore, the addition of GCs is necessary to improve the conductivity of IC/LFP.

The morphology and structure of the LFP materials are characterized by SEM and TEM (Fig. 6). It is confirmed that the phytate is the result of secondary crystallization after hydrothermal reaction (Fig. 6a). At this moment, the crystals take on spheroidicity with diameter about 1 μm , each of which is composed of plenty of tablet-like nanocrystals with diameter around 50 nm (inset in Fig. 6a). These tiny building blocks are assembled side by side in an ordered fashion. As seen in Fig. 6b, the phytate crystals take on square and sheet-like structure. The crystals are arranged uniformly and fitly, which is in agreement with the corresponding SEM images. However, the morphology and structure of the precursor is thoroughly changed after calcining process. As seen in Fig. 6c, the structure of crystals is changed into disordered and cluttered bulk crystals. The resulted phosphates present bulk crystals with edges and corners, which is different from GC/LFP crystals (Fig. 6e) by using PA. The common LFP crystals are smooth and mellow. Fig. 6c shows that the diameter of our bulk crystals is ranged from 100 to 300 nm. The precursor crystals in previous reports [23,33] are large and fused into the small final crystals, while the precursor crystals, in our work, are small and aggregated into large crystals during calcining process. In addition, the ICs in the IC/LFP cannot be observed through SEM (Fig. 6d). From what has been discussed above, the IC sheets are separated from the phytates. It is speculated that the ICs exist inside the LFP crystals and the GCs coat the LFP crystals outside (Fig. 6f). The high resolution TEM of GC/IC/LFP (Fig. S4) shows the IC/LFP particles are coated with the GC coating and the thickness of the GC layer is about 2.5 nm.

The comparison of crystalline nature between the GC/LFP and GC/IC/LFP is conducted by XRD as shown in Fig. 7. It is found that almost all the peaks of GC/LFP conform to those of GC/IC/LFP (Fig. 7a), indicating that the phytates are decomposed into LFP during the high temperature calcination process. However, via local amplification of the XRD patterns between 25° and 36.5° [(111), (211), (020), (301) and (311) crystal planes of LFP, Fig. 7b], it is obvious that all the peaks of GC/IC/LFP have shifted to the left about 0.25° because of the existence of ICs with the oxygenic groups. It would lead to the increase of some lattice parameters. The full width half maximum (FWHM) of the XRD peaks in Fig. 7 has also been quantified. The value of FWHM of (311) crystal plane for GC/IC/LFP is 0.240° , which is larger than that for GC/LFP (0.214°). It is further proved that GC/IC/LFP has a wider distribution of lattice constants due to the existence of the internal carbons within LFP particles. In addition, the XRD patterns for LFP, IC/LFP and GC/LFP have been collected (Fig. S5) and the Williamson-Hall analysis have been conducted to quantify the amount of strain in the composites (Fig. S6). The amount of residual strain in IC/LFP is 0.112, larger than those in GC/LFP (0.049) and LFP (0.022). The larger residual strain may be caused by some IC sheets lie out of LFP lattices. As seen in Fig. 7c and d, both of GC/IC/LFP and GC/LFP are highly ordered single crystals. However, the crystalline interplanar spacing of GC/IC/LFP is slightly larger than that of GC/LFP, which is corresponding to the result of XRD patterns. It would be beneficial to the transportation of lithium ions, which is a reason for the good performance. It is noteworthy that Wei *et al.* [23] doped the triphenylphosphine into GC/LFP and acquired an excellent electrochemical performance. Especially, after the triphenylphosphine was doped, the peaks of XRD pattern of composite slightly shifted to left. It is further illustrated that the carbons, derived from triphenylphosphine or phytic acid, can influence lattice parameters of LFP, which is different from the carbons derived from glucose.

In order to investigate the electrochemical performance of the electrodes, the cyclic voltammetry (CV) test is shown in Fig. 8. The CV of GC/LFP (black line) shows a pair of anodic/cathodic peaks at about 3.62 and 3.3 V, respectively, while CV of GC/IC/LFP (red line) shows

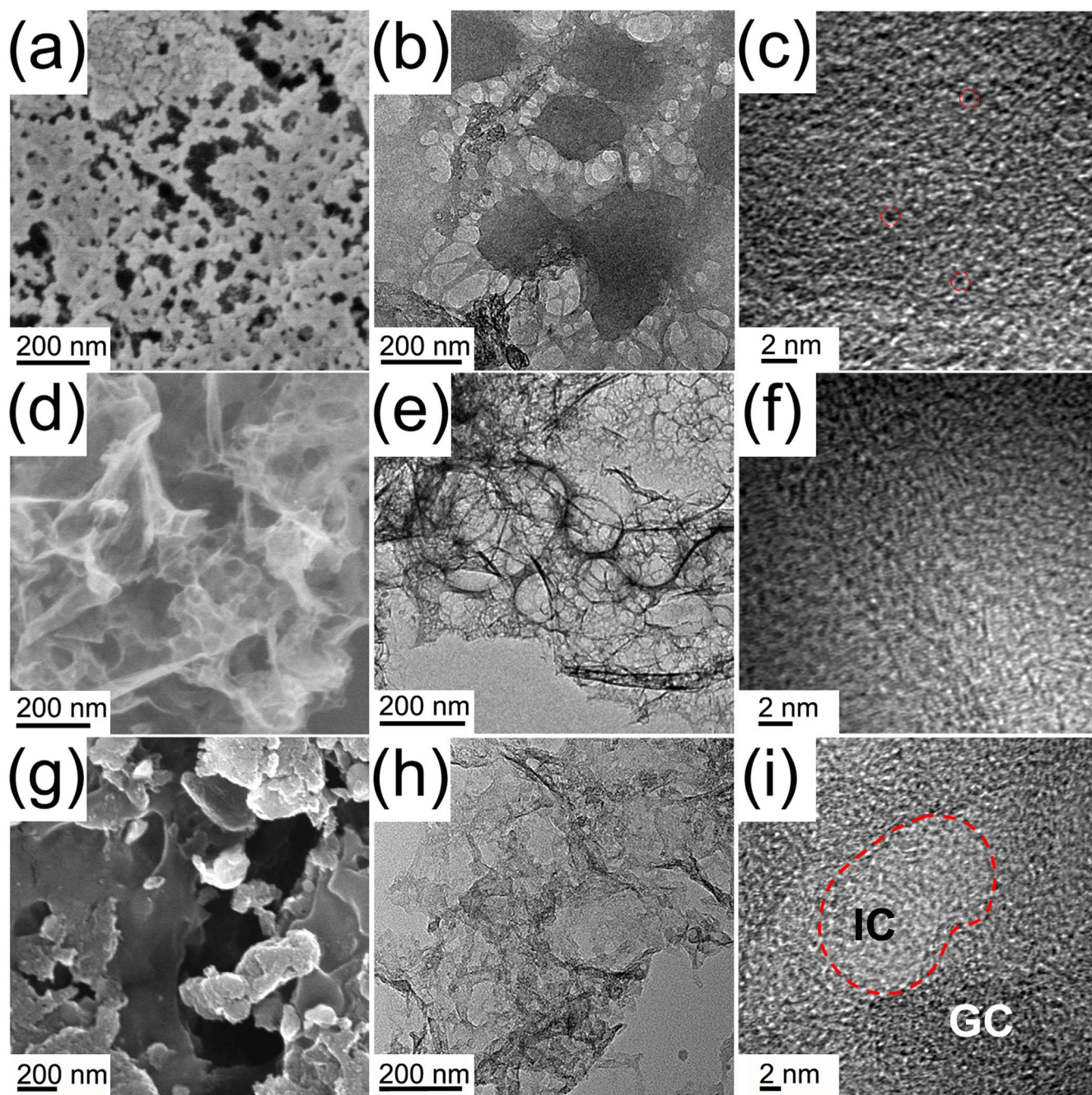


Fig. 5. SEM (a), TEM (b) and HRTEM (c) images of ICs, SEM (d) TEM (e) and HRTEM (f) images of GCs, and SEM (g) TEM (h) and HRTEM (i) images of GC/IC.

two pairs of anodic/cathodic peaks. The area of CV with the secondary anodic/cathodic peaks for GC/IC/LFP is larger than that for GC/LFP, indicating that GC/IC/LFP has a higher capacity than GC/LFP. Thereinto, the main anodic/cathodic peaks are at about 3.51 and 3.33 V, which is consistent with the two-phase redox reaction of LFP. The potential separation of main anodic/cathodic peaks is 0.18 V, which is much lower than that (0.32 V) of GC/LFP. It is demonstrated that GC/IC/LFP possesses a better ionic conductivity due to the effect of ICs. What's more, the secondary anodic/cathodic peaks are at about 3.59 and 3.24 V (red arrow), respectively. And the additional anodic/cathodic peaks are corresponded to the excess capacity, which can be confirmed by the derivative of galvanostatic charge–discharge curve below.

Compared with the C/LFP/ECGO electrode [33], as reported in one of our previous work by using PA, the GC/IC/LFP electrode displays a higher specific capacity. The galvanostatic charge–discharge curve of

C/LFP/ECGO at current rate of 0.1 C is shown (the black line in Fig. 9a), which displays a high specific capacity. The corresponding differential curve shows only one pair of peaks (the black line in Fig. 9b). On the contrast, the specific capacity of GC/IC/LFP, with the optimal ratio, comes to 192 mA h g^{-1} at current rate of 0.1 C (the red line in Fig. 9a), which exceeds the theoretical capacity of LFP, 170 mA h g^{-1} . The differential curve of galvanostatic charge–discharge profile is showed (the red line in Fig. 9b). It is clear that there exist two pairs of peaks in the charge–discharge curve, which illustrates that the large capacity is attributed to other sources except for the contribution of LFP. The result conforms to the corresponding cyclic voltammograms (Fig. 8).

The reason that the capacity of GC/IC/LFP exceeds the theoretical capacity of LFP can be mainly addressed to the IC sheets. As discussed above, some O-containing groups such as —C=O and —COOH exist on the IC sheets. For carbon material, the introduction of surface O-

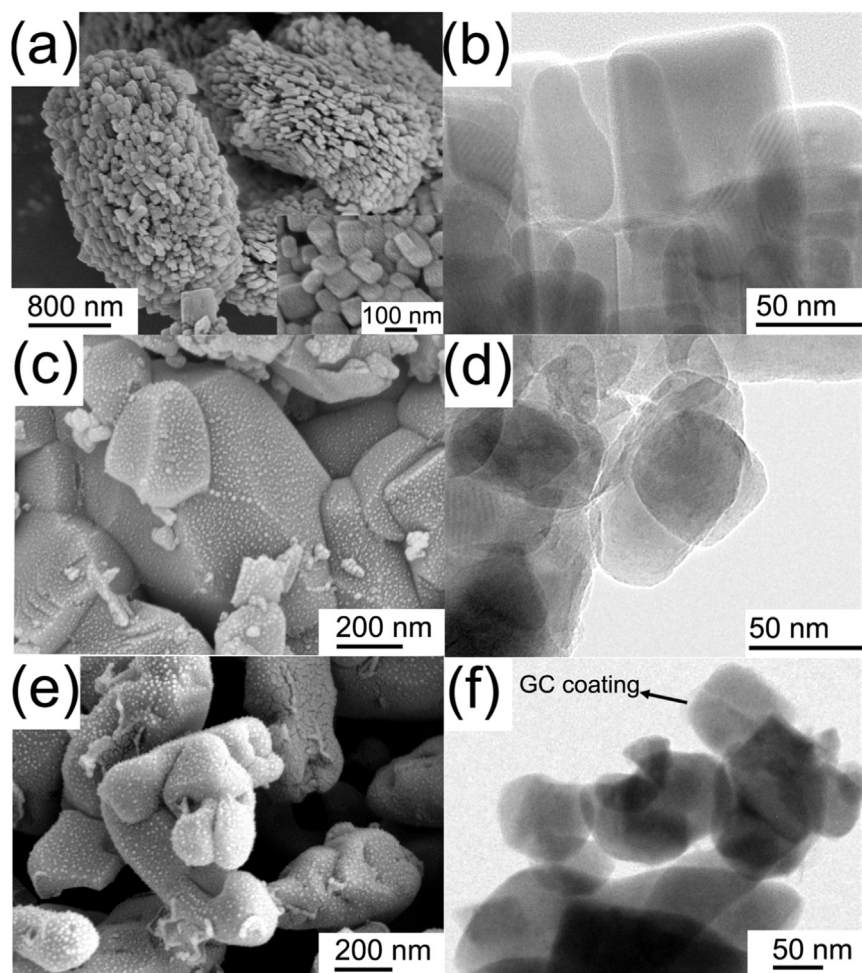


Fig. 6. (a and b) SEM and TEM of the precursor of IC/LFP, (c and d) SEM and TEM of IC/LFP, (e and f) SEM and TEM of common GC/LFP.

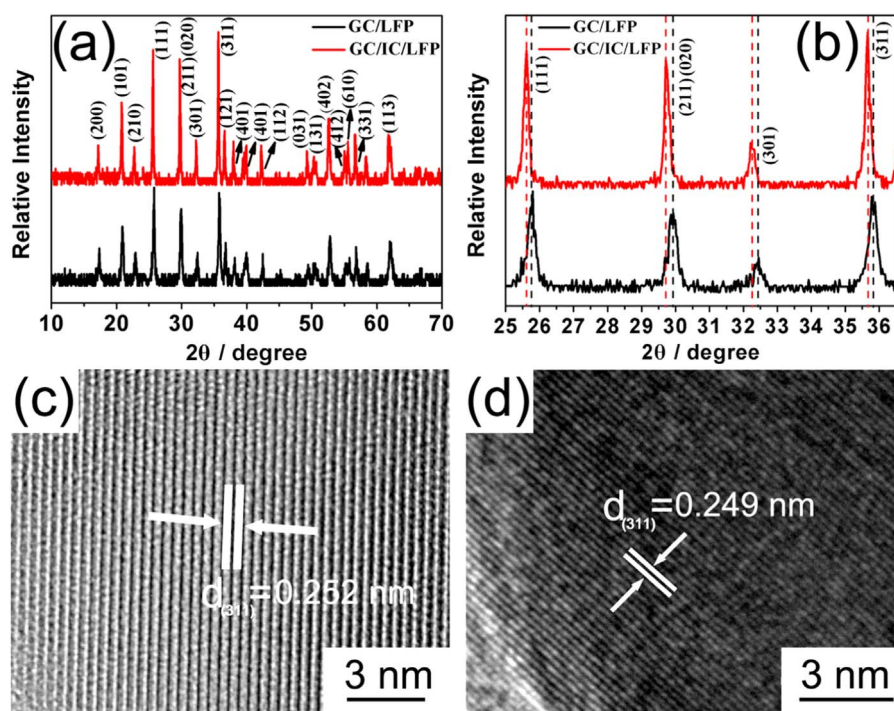


Fig. 7. (a) XRD pattern of GC/LFP and GC/IC/LFP, (b) local amplification of XRD patterns corresponding to the strong peaks, and HRTEM images of GC/IC/LFP (c) and GC/LFP (d).

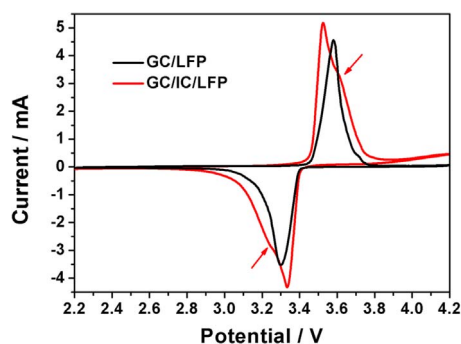


Fig. 8. Cyclic voltammograms of GC/LFP and GC/IC/LFP at a scan rate of 0.1 mV s⁻¹.

containing functional groups, is a promising method to increase the energy density of electrodes. These oxygen atoms in the carbonyl groups and various other oxygen groups usually bound to the edges of carbons [48,49]. These surface oxygen functional groups can undergo Faradaic reactions, while the characteristic reversible Faradaic peaks locate at around 3.0 V vs. Li and 3.4 V vs. Li [47]. The redox peaks can be assigned primarily to the reaction: $\text{-C=O} + \text{Li}^+ + \text{e}^- \leftrightarrow \text{-C-OLi}$ [50,51]. And the position of redox peaks of oxygen groups will be varied by different content of carbon. Hye et al. [47] have fabricated a functionalized multiwalled carbon nanotube doped with graphene oxide, with this structure, its redox peaks appeared at around 2.3 and 3.1 V vs. Li. The CVs of ICs and GCs electrodes were shown as Fig. S7. The redox peaks of the oxygen groups on IC sheets were observed at ~3.3 and 3.5 V, which no obvious peaks were done for GCs with few O-containing groups. To investigate directly the lithium storage performance of ICs and GCs, the IC/SiO₂ and the GC/SiO₂ cathodes were prepared. The carbon content in IC/SiO₂ or GC/SiO₂ cathode materials is 2%, same as that in IC/LFP or GC/LFP. The cathodes were also assembled into the button cells in the same method of GC/IC/LFP. Fig. 9c shows the galvanostatic charge–discharge curves of bare SiO₂, GC/SiO₂ and IC/SiO₂ at current rate of 0.1 C. As shown in Fig. 9c, the bare SiO₂

Table 1

The ratios of Li to Fe to P for GC/LFP and GC/IC/LFP.

Composite	Pristine synthesized material	Charged electrode	Discharged electrode
GC/LFP	1.07: 1: 1.02	0.22: 1: 1.04	1.02: 1: 1.01
GC/IC/LFP	1.35: 1: 1.08	0.18: 1: 1.06	1.29: 1: 1.08

barely provides capacity while GC/SiO₂ delivers a small reversible capacity. It is proved that the GCs can improve the conductivity of material but cannot provide the extra capacity. However, the capacity of IC/SiO₂ nearly reaches 35 mA h g⁻¹, which is consistent with the excessive capacity observed in Fig. 9a. The differential curve of galvanostatic charge–discharge profile of IC/SiO₂ was shown in Fig. 9d. It is found that the redox potentials lie at about 3.35 V and 3.49 V, which are consistent with the CV curve of ICs (Fig. S7) and correspond to the secondary redox peaks in Fig. 9b.

From the point of energy conservation, the extra capacity has to be related to the excess of Li⁺. Based on the speculation, a careful chemical analysis has been conducted for GC/LFP and GC/IC/LFP to quantify the ratio of Li to Fe to P elements. The amount of Li and Fe for GC/LFP and GC/IC/LFP were determined by the ICP-MS analysis while the amount of P was determined by the Phosphomolybdic acid precipitation method. The results are shown in Table 1, when phosphoric acid is served as the phosphorus source in the preparation step, the ratio of Li to Fe to P is approximately 1:1:1. After being charged, about 80% of lithium was transferred to the anode. And the vast majority of the transferred Li⁺ could return when discharging. However, as a comparison, the actual Li-doping amount is increased obviously by using phytic acid when synthesizing cathode material, which the ratio of Li to Fe to P reaches 1.35:1:1.08, indicating the excess amount of Li and P. It can be found that more Li⁺ can be transferred to the anode after being charged and the majority of them can return when discharged. It has been reported that the excess lithium can be involved in the reversible redox reaction. The excess Li and P could bind with the oxygenic

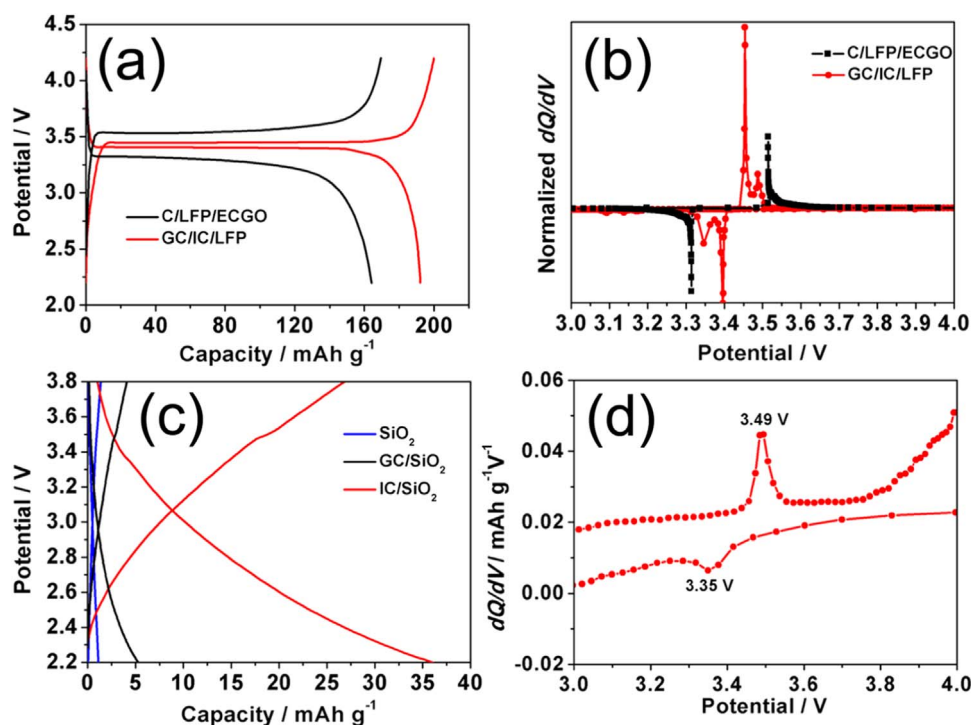


Fig. 9. (a) Galvanostatic charge–discharge curve of C/LFP/ECGO and GC/IC/LFP at current rate of 0.1 C; (b) the differential curves of the galvanostatic charge–discharge profiles of C/LFP/ECGO and GC/IC/LFP; (c) galvanostatic charge–discharge curve of SiO₂, GC/SiO₂ and IC/SiO₂ at current rate of 0.1 C; and (d) the differential curve of the galvanostatic charge–discharge profile of IC/SiO₂.

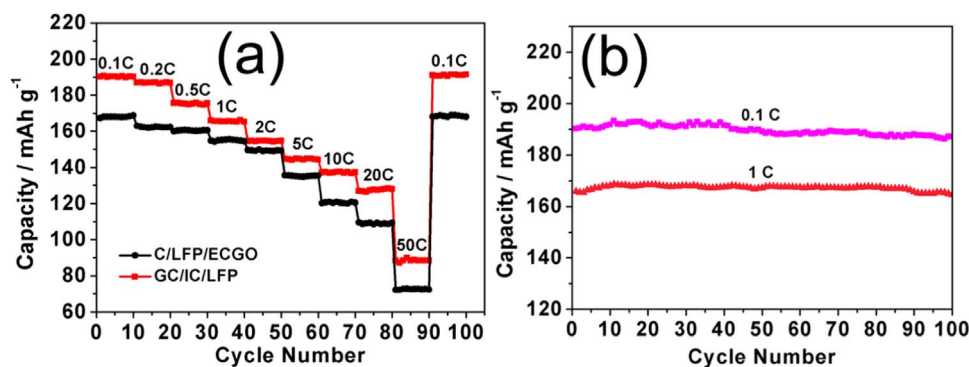


Fig. 10. (a) Discharge cycling performances of GC/IC/LFP and C/LFP/ECGO at different discharge rates from 0.1 to 50 C, and (b) cycling capability of GC/IC/LFP at current rates of 0.1 C and 1 C.

groups or some defects on the IC sheets, while the oxygenic group bind by lithium can undergo reversible Faradaic reactions, which explains the reason for the excess capacity.

Furthermore, the ratios of reactants and the washing condition would affect the ratios of Li to Fe to P, which would affect further the final electrochemical performance of GC/IC/LFP. Therefore, the optimization for the ratio of reactants and the washing condition is necessary and has been performed (Figs. S8 and S9). In this paper, all of the materials are prepared under the optimal condition.

The specific capacities of GC/IC/LFP and C/LFP/ECGO electrodes are tested at rates of 0.1 C, 0.2 C, 0.5 C, 1 C, 2 C, 5 C, 10 C, 20 C and 50 C, which are charged and discharged for 10 cycles at each rate, and then tested at 0.1 C for another 10 cycles (Fig. 10a). It is clear that the GC/IC/LFP electrode has a better rate performance, higher capacity and better cycling stability than C/LFP/ECGO. The specific capacity can reach 140 mA h g⁻¹ even at 10 C. Almost complete recovery of the capacity can be attained when the rate is decreased to 0.1 C for the GC/IC/LFP electrode, suggesting that the electrode, along with the formed solid electrolyte interface (SEI), is quite stable during cycling. The rate performances for GC/IC/LFP have been compared with those of the state-of-art LFP-composites reported recently [13,33,52–57]. It can be found that even at the higher current rates of 20 C and 50 C, the GC/IC/LFP electrodes still deliver higher capacities, exhibiting a prominent rate capability. Except for the current densities, the loading mass of the electrode have influences on the capacities. Therefore, the effect of loading mass of the prepared electrode on the capacity needs to be discussed. As shown in Fig. S10, the capacity for GC/IC/LFP decrease slightly with the increase in the loading mass due to the increased polarization [58,59]. In this paper, the mass loading of electrodes is about 2.8 mg cm⁻². When a higher mass loading of 4.2 mg cm⁻² is employed, the capacity for GC/IC/LFP can also reach to 187 mA h g⁻¹, higher than theoretical capacity for LFP at 0.1 C. Simultaneously, the capacity for GC/IC/LFP with a relatively small mass loading (1.4 mg cm⁻²) can reach 200 mA h g⁻¹ at 0.1 C. The long-term cycling performance of the GC/IC/LFP electrode at current rates of 0.1 C and 1 C is shown in Fig. 10b. It is clear that the capacity losses at different conditions are negligible at the 100th cycle.

Fig. 11a and b show the specific discharge capacity, at 0.1 C, of GC/IC/LFP and C/LFP/ECGO at various charge currents. It is found that for GC/IC/LFP, the specific discharge capacities have an insignificant decrease with the increase of charge current. The specific charge capacity can reach 150 mA h g⁻¹ at 2 C within 30 min while the specific discharge capacity maintains at 186 mA h g⁻¹. The performance of C/LFP/ECGO is analogous to that of GC/IC/LFP. The only difference between these two materials in performance is that the specific discharge capacity of C/LFP/ECGO decreases obviously with the increase of charge current. In particular, the specific charge capacity can reach 135 mA h g⁻¹ at 2 C and the specific discharge capacity just remains at 154 mA h g⁻¹ at 0.1 C.

From Fig. 11a and b, the discharge capacity of GC/IC/LFP or C/LFP/ECGO, at 0.1 C, is much larger than the charge capacity at large current. It seems that the performances of GC/IC/LFP and C/LFP/ECGO disobey the law of energy conservation, which verified below. As shown in Fig. 11c, d and e, the different color areas, surrounded by the charge or discharge curve, represents the charge and discharge energy density, respectively. The size of the areas can be calculated by means of the integral of charge (Eqs. (1) and (2)) and discharge curves (Eq. (3)). The energy efficiencies of GC/IC/LFP and C/LFP/ECGO can be compared by calculating the ratio between discharge energy density and charge energy density.

$$w_1 = \int_0^t V(t) \cdot Idt \quad (1)$$

$$w_2 = \int_0^t V \cdot I(t) dt \quad (2)$$

$$w_3 = \int_0^t V(t) \cdot (-I) dt = - \int_0^t V(t) \cdot Idt \quad (3)$$

Where three equations stand for the energy of constant current charge (w_1), constant voltage charge (w_2) and constant current discharge (w_3). Besides, V represents the voltage, I represents the charge current, $-I$ represents the discharge current and t represents the charge and discharge time.

The corresponding areas were integrated according to the equations above. Based on the same mass loading, the charge and discharge energy can also be calculated. As shown in Fig. 11c, the charge energy of GC/IC/LFP at 0.1 C is 2656 mWh and the charge energy at 4.2 V is 35 mWh. The total amount of charge energy (2591 mWh) is slightly larger than the discharge energy (2477 mWh) at 0.1 C, which accords with energy conservation except a small deviation. The energy efficiency can reach 96%. As seen in Fig. 11d, the charge energy of GC/IC/LFP at 2 C is 2150 mWh while the charge energy at 4.2 V is 466 mWh, the total amount (2616 mWh) of which is also larger than the discharge energy (2354 mWh) at 0.1 C. The energy efficiency of GC/IC/LFP reaches 90%. However, for the C/LFP/ECGO, the charge energy at 2 C is 1800 mWh while the charge energy at 4.2 V is 615 mWh. And the total charge energy (2415 mWh) largely exceeds the discharge energy at 0.1 C (1956 mWh). The energy efficiency of C/LFP/ECGO is only 81%. Therefore, the energy efficiency of GC/IC/LFP is superior to that of C/LFP/ECGO. Some capacity losses like the high resistance caused electro-thermal conversion during the charging-discharging process can result in the low energy efficiency of C/LFP/ECGO. Consequently, it is speculated that the IC sheets play a key role in extracting Li ions in charging process. It is proved that the GC/IC/LFP can more fully turn the charge capacity into the discharge capacity.

Fig. 12 is an illustration of charge process for GC/IC/LFP. The red globules on ICs in Fig. 12 represent the oxygenic groups, such as carboxyl groups. The oxygenic groups can release and capture Li⁺ in the charge-discharge process. The excessive Li⁺ are stored reversibly

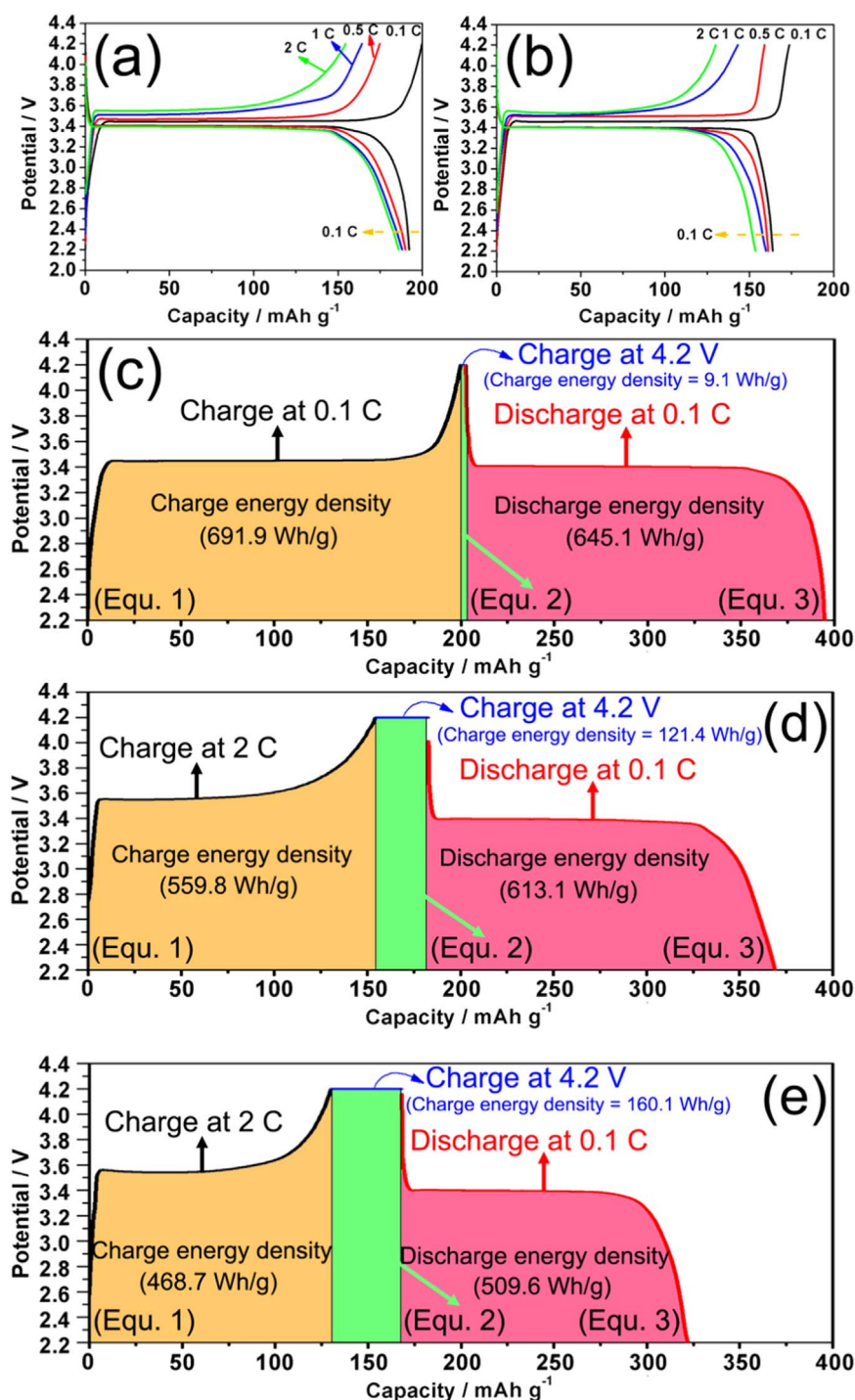


Fig. 11. Galvanostatic charge–discharge curves of GC/IC/LFP (a) and C/LFP/ECGO (b) at various charge and discharge rates. The energy efficiency illustration of GC/IC/LFP (c) at 0.1 C charge current and 0.1 C discharge current, and the energy efficiency illustration of GC/IC/LFP (d) and C/LFP/ECGO (e) at 2 C charge current and 0.1 C discharge current.

between IC sheets or the defects and vacancies at edge sites. As discussed above, the IC sheets with phosphate groups are doped into LFP crystal, which leads to the increasing of interplanar spacing of the crystals. It is speculated that the existence of ICs can promote Li ions transportation in LFP crystals. In the charge process, a majority of lithium ions are extracted according to the normal way (the blue dotted line) while the other part of lithium ions can be extracted through the IC sheets (the yellow dotted line). As shown in Fig. 12, the lithium ions on the defects and edges of IC sheets are first extracted (the purple dotted line). The remaining oxygenic groups in IC sheets provide many binding sites for the lithium ions of LFP. Some lithium ions pass

through the IC sheets. Therefore, the lithium ions can be easier to be extracted under the large current charging with the help of IC sheets. The two modes of lithium ions' extraction are corresponding to the two oxidation peaks in the cyclic voltammetric curve for GC/IC/LFP (Fig. 8). From the result, it is also proved that the ionic conductivity of GC/IC/LFP is higher than that of the common LFP, which agrees the result of EIS (Fig. S11 in the ESI[†]). The lithium diffusion coefficients (D) for GC/IC/LFP and GC/LFP have been calculated and compared. The values of D are calculated from the inclined lines in the Warburg region using the following equation:

$$D = R^2 T^2 / 2 A^2 n^4 F^4 C^2 \sigma^2 \quad (4)$$

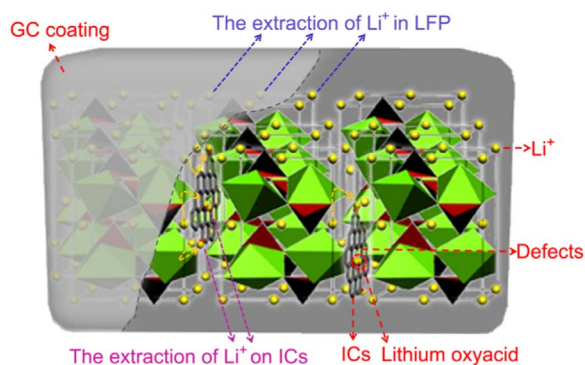


Fig. 12. The anatomical structure illustration of GC/IC/LFP in charge process.

or where R is the gas constant, T is the absolute temperature, A is the surface area of the cathode electrode, n is the number of electrons per molecule during oxidation, F is the Faraday constant, C is the Li^+ concentration, and σ is the Warburg factor associated with Z' ($Z' \propto \sigma \omega^{-1/2}$). The apparent Li diffusion coefficient D could be calculated from the above equation after linear fitting the relation plot between Z' and the reciprocal square root of the angular frequency ω (Fig. S12) to estimate the Warburg factor σ . The D values for GC/LFP and GC/IC/LFP were calculated to be $1.13 \times 10^{-14} \text{ cm}^2 \text{ s}^{-1}$ and $2.47 \times 10^{-13} \text{ cm}^2 \text{ s}^{-1}$, respectively. It is clear that GC/IC/LFP has a larger D value than GC/LFP, which indicates that the existence of ICs is more favorable for promoting the Li diffusion processes. The reasons for this can be concluded as follows: (i) the IC sheets can enlarge the lattice spacing of LFP which benefits intercalation and deintercalation of Li^+ ; (ii) the IC thin sheets themselves can be used as bridges to provide lots of pathways for Li-ion migration and electrolyte diffusion. The presence of IC sheets in LFP can reduce the potential barrier between the electrolyte and the crystals, and promote the transportation of Li^+ . When it is charging, Li^+ will firstly get off from LiFePO_4 lattices through the (010) axis, and then migrate along the IC sheets. When discharging, the reverse happens.

An additional explanation is also necessary that the glucose doping is needed when testing the performance of cathode material. The typical bulk resistivity of IC/LFP is about $2 \text{ k}\Omega \text{ cm}$ measured by the four point probe tester with a measurement range from 10^{-4} to $10^5 \Omega \text{ cm}$. As a result, the typical bulk resistivity of GC/IC/LFP is about $80 \Omega \text{ cm}$, which is superior to that of GC/LFP, about $140 \Omega \text{ cm}$. It is also an advantage to make GC/IC/LFP exceed the theoretical capacity. Moreover, the electronic conductivity of GC/IC/LFP is better than that of GC/LFP, which can be demonstrated by their EIS spectra (Fig. S11 in the ESI†).

Nowadays, lithium cobalt oxide (LiCoO_2) is widely employed in all kinds of portable sets, such as phones, computers and cameras. The major advantage of LiCoO_2 is the high volume energy density (1200 Wh L^{-1}) due to its high tap density (2.2 g mL^{-1}), which is almost twice as high as that of commercial LFP. However, the major disadvantage of LiCoO_2 is capacity fading, which lead to the fact that the cycle number of LiCoO_2 is just 500 cycles. Therefore, a big problem for commercial LiCoO_2 battery is lifetime. When it comes to GC/IC/LFP, the specific capacity increases to 192 mA h g^{-1} at 0.1 C . In our previous work, the C/LFP/ECGO composite was prepared by an in situ solvothermal approach. The LFP can grow in-situ on the electrochemically cleaved graphite oxides combined with a cold-press calcination process. Both of the electrochemically cleaved graphite oxides and the glucose-derived carbon coating, produce an effective three-dimensional (3D) conducting network. The tap density could increase to 1.5 g mL^{-1} by this way. Therefore, the tap density of GC/IC/LFP is expected to increase by the method in the following work. Both of the high specific capacity and the high tap density make the volume energy density of GC/IC/LFP almost reach 1000 Wh L^{-1} . The volume energy density of

GC/IC/LFP would be increased by 67% compared with that of common commercial LFP (about 600 Wh L^{-1}). In addition, the lifetime of commercial LFP can achieve 2000 cycles, which greatly exceeds that of LiCoO_2 . Therefore, it is highly possible that GC/IC/LFP would be a promising material in the application of some portable devices in consideration of the high safety, high volume energy densities and long lifetime. It is found that the specific capacity of GC/IC/LFP reaches 190 mA h g^{-1} at different charge rates, which exhibits the excellent capacity retentions for the GC/IC/LFP electrodes at high charge rates. Therefore, if the GC/IC/LFP is applied in the cellphones, the standby time of cellphones would be extended under the same charging current and time in that the discharge capacity is higher than that of the common LFP or LiCoO_2 . Consequently, the GC/IC/LFP is possible to be applied in the portable devices.

4. Conclusions

In this work, a carbon-encapsulated lithium iron phosphate with internal carbon sheets composite, GC/IC/LFP, is synthesized by employing PhyA as phosphorus source with the hydrothermal-alcohol washing-calcination procedure. The precursor is composed of phytate spheroid particles with tablet-like nanocrystals, which are self-assembled in an ordered fashion. The submicrometer-sized bulk materials are formed after calcining. It is found that a little amount of internal carbons, ICs, could increase the ionic conductivity of GC/IC/LFP. The electrochemical performances of GC/IC/LFP, such as stability and ability for capacity recovery, are better than those of common LFP prepared using PA, even though with a high content of active material in working electrodes (85%). Especially, the as-prepared GC/IC/LFP shows an ultra-high specific capacity of 192 mA h g^{-1} at 0.1 C , which exceeds the theoretical capacity of LFP, with a remarkable rate performance of 140 mA h g^{-1} at 10 C . The energy efficiency of GC/IC/LFP reaches nearly 96% at 0.1 C . It is speculated that the extra capacity is mainly ascribed to the reversible redox reaction between the lithium ions of the electrolyte and the oxygenic groups at defects in ICs. The increasement of ionic conductivity for GC/IC/LFP can be attributed to the following factors: (i) the internal carbons, separated from PhyA, improve the transportation of lithium ions, (ii) the interplanar spacing of IC/LFP bulk crystals, with edges and corners, is larger than common LFP, which would effectively facilitate lithium ions diffusion, and (iii) the stable structure acquired in our specific synthetic way. Finally, the preparation method and mechanism explanation improve the application of LFP. Furthermore, it is anticipated that the GC/IC/LFP shows the potential possibility of application in the portable sets.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nanoen.2017.03.006.

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